each other in different ways.

APPLICATION SERVICES

Application services are stateless and are responsible for implementing any kind of logic. They don’t have to obey specific rules like those for microservices, as long as they expose all the cloud native properties you learned about earlier in this chapter.

It’s paramount that you design each service with loose coupling and high cohesion in mind. Services should be as independent as possible. Distributed systems are complex, so you should be extra careful during the design phase. Increasing the number of services leads to an increased number of problems.

You’ll probably develop and maintain most of the application services in your system by yourself, but you can also use some offered by cloud providers, such as authentication or payment services.

DATA SERVICES

Data services are stateful and are responsible for storing any kind of state. The *state* is everything that should be preserved when shutting down a service or spinning up a new instance.

Data services can be relational databases like PostgreSQL, key/value stores like Redis, or message brokers like RabbitMQ. You can manage these services by yourself. Doing so is more challenging than managing cloud native applications due to the storage required to save the state, but you will gain more control over your own data. Another option is to use the data services offered by a cloud provider, in which case the provider will be responsible for managing all the concerns related to storage, resilience,